Optimisation Workflow Manual

Overview

This guide provides a comprehensive, step-by-step workflow for using the **iolite Optimiser**. It combines the **Single Pulse Response (SPR)** analysis with the **Optimiser** to ensure your laser ablation parameters are perfectly tuned for your specific hardware and analytical goals.

Phase 1: Determine Washout (SPR Tab)

Before optimising spot size or scan speed, you must characterise your system's Single Pulse Response (SPR) to determine the washout time.

Step 1.1: Acquire and Load SPR Test Data

1. **Acquire Test Data**: Run a single-pulse test ablaton on your instrument using a low laser repetition rate of **1 Hz** and a **10 ms** integration time measuring a single mass (isotope). This ensures you record cleanly separated, high-resolution profiles of individual laser pulses.
2. Open the **iolite Optimiser** plugin from the *Tools* menu in iolite.
3. Go to the **SPR Tab**.
4. Click **Import SPR File** and select your SPR data file.

Step 1.2: Select & Optimise Peak Detection

1. **Select Channel**: Choose the high-abundance isotope channel (e.g. `U238`, `Th232`) used during the single-pulse run.
2. **Verify Detection**: Look at the **SPR Plot**. Valid pulses should be marked with an orange `X`.
    - *Too much noise?* Increase **Peak Cutoff** (uncheck Auto) or **Min. Distance**.
    - *Missed peaks?* Decrease **Peak Cutoff**.
3. **Clean Data**: If you see double-peaks or artifacts, click on them in the plot to **Exclude** them from the calculation.
4. **Apply Washout**:
    - Review the **FW 0.1M (10%)** statistics in the left panel.
    - Click **Apply Average, Max, or Composite to Optimiser**.
    - *Tip*: The **Composite** value is often the most stable representative of your system's washout.

Phase 2: Determine PreScan Parameters (Setting Initial Conditions)

Once the Single Pulse Response washout is determined, you can use the advisor tool to set up the initial conditions for your instrument PreScan.

Step 2.1: Calculate Suggestions

1. Switch to the **Optimiser Tab** of the plugin.
2. In the **Input Parameters** group:
3. Enter your target **Initial Spot Size** (e.g. 15μm) and the **Washout** decay time applied from Phase 1.
4. Enter the **Number of Analytes** representing the total number of isotopes in your proposed mass spectrometer method.
5. Click the **Suggest PreScan Params** button.
6. The suggestions dialog will display:
7. **Dwell Time per Isotope**: Bounded by hardware minimums and rounded to the instrument's resolution.
8. **Target Cycle Time**: Overall integration loop time.
9. **Suggested Rep-Rate**: Firing frequency that achieves low statistical fluctuation ($\leq 5\%$ RSD) under steady-state oversampling.
10. **Suggested Stage Speed** and **Overlap**: Recommended values to prevent striping.
11. Apply these settings to your instrument's control software to execute the PreScan and generate the primary calibration/tuning data file.

Phase 3: Optimiser Setup (Optimiser Tab)

Step 3.1: Hardware Configuration

1. Switch to the **Optimiser Tab**.
2. Click the **Settings** button.
3. **ICP-MS**: Select your Manufacturer and Model.
   - *Custom?* Choose "Custom Model" and enter your **Min Dwell Time** and **Dwell Resolution** (e.g., for TOF or generic Quad).
4. **Laser**: Select your Platform and Cell Type.
   - *Custom?* Choose "Custom Laser" and manually limit the **Max Rep Rate**, **Max Speed**, and **Rep-Rate Resolution**.

Step 3.2: Load Sensitivity Data

1. Ensure your iolite session has a representative ablation signal (e.g., a "Tuning" line or standard block).
2. Click **Import Optimisation File** and select your data file.
3. **Dwell Time Check**:
   - If your data lacks dwell time metadata, a dialog will appear.
   - Choose **Set to Global** (e.g., 10ms) or enter **Individual** times per channel to match your instrument method.
4. **Region Selection**:
   - Check the Signal Plot.
   - Ensure the **Background (Blue)** region covers the gas blank.
   - Ensure the **Signal (Red)** region covers the steady ablation signal.
   - **Quick Adjustment**: Hold **Shift** and drag the region lines directly on the plot. The software will automatically swap start/end points if you cross them.
   - Use **Auto Select Regions** if they are incorrect.

Phase 4: Run Optimization

Step 4.1: Define Analytical Goals

1. **Mode**: Select your analysis type (usually **Imaging** or **Line**).
2. **Washout**: Verify this matches the value you applied from the SPR tab (Phase 1).
3. **Parameters**:
    - **Sync Strategy**: Choose how rep-rates and integration times are optimised:
    - **Adaptive Integer Sync (Auto)**: Sweeps frequencies to guarantee integer pulses per cycle.
    - **Strict Pulse Target (Manual)**: Hard-locks to the target pulses.
    - **Oversampling (Time-Driven)**: Prioritizes steady-state oversampling without snapping.
    - **Combined Integer Sync (Auto)**: Sweeps above the oversampling minimum to find a perfect integer sync count.
    - **Sync Target (Pulses)**: Set the strict integration synchronisation requirement (e.g., 10 shots/dwell).
    - **Sync Target (RSD %)**: Set the target stability margin (e.g. 5%) under oversampling modes.
    - **Prefer Exact Pulses per Acq**: Check this box to force the algorithm to search only for the exact integer target.
    - **Dosage**: Set the spatial density requirement (e.g., 10 shots/pixel).
        - Keep **Sync Dosage** checked for square pixels where Dosage = Pulses/Dwell.
        - Uncheck **Sync Dosage** to manually decouple the scan speed from the mass spectrometer's integration rate (e.g., for rectangular pixels or TOF instruments).
    - **Target SNR**: Set the desired quality (e.g., 10 Sigma).
    - **Avoid Gaps**: Check this for Imaging to ensure full surface coverage.
4. **Navigation Tip**: **Double-click** any plot at any time to instantly reset the view to its full extents. Use the **Mouse Wheel** to zoom and **Left-Click & Drag** to pan.

Step 4.2: Interpret Results

The software calculates the optimal settings automatically.

1. **Review Settings**: Look at the **Optimised Settings** panel for your ideal **Spot Size**, **Rep-Rate**, and **Speed**.
2. **Review Synchronization Advisor**:
3. Check the advisor diagnostics in the results summary. Ensure it displays green icons indicating **Synchronised**, **Oversampling Mode**, or **Stable Steady State** (all channels meeting target RSD). If a yellow/red status warning appears, adjust dwell or rep-rates accordingly.
4. **Check Constraints**: Look at the **Results Table**.
    - **Orange Rows**: These isotopes cannot meet the Target SNR even at the maximum allowable settings.
    - **Blue Rows**: These are limited by your hardware speed (Min Dwell).
5. **Refine**:
    - *Spot Size too big?* Reduce the **Target SNR** or **Sync Target (Pulses)**.

    - *Scan too slow?* Uncheck **Avoid Gaps** (if acceptable) or increase **Spot Size** manually.

Step 4.3: Manual Overrides

- **Force Spot Size**: If you strictly need a specific beam diameter (e.g., 25µm), enter it in the **Spot Size** spinbox (toggle off "Auto"). Valid Rep-Rate and Speed will be re-calculated for that fixed spot.